SELLER CONCENTRATION & FREIGHT RISK ANALYSIS

Seller Concentration, Freight Risk & Cross-Sell Analysis of a 100K-Order E-Commerce Marketplace

Business Conclusion Report

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August 2026

Executive Summary

This report analyzes 99,441 orders, 3,095 sellers, and 96,096 unique customers on a Brazilian e-commerce marketplace over 2016-2018. Rather than repeating the standard "late delivery hurts review score" narrative common to this dataset, the analysis targets three underexplored commercial risks: how concentrated marketplace revenue is among a small group of sellers, how much cross-state logistics inefficiency is costing the platform, and how little cross-selling is actually happening across product categories.

Marketplace Snapshot

KPI	Value	What it tells us
Total Revenue	$13.41M	Item-level revenue across all orders
Total Orders	99,441	Orders placed 2016–2018
Total Customers	96,096	Unique customers (customer_unique_id)
Total Sellers	3,095	Active sellers on the platform
Average Order Value	$134.84	Revenue ÷ total orders
Delivered Order Rate	97.04%	Share of orders successfully delivered
Top 10 Seller Revenue Share	13.15%	Revenue concentration in 10 sellers
Top Decile Seller Revenue Share	67.5%	Revenue concentration in top 10% of sellers
Total Freight Cost	$2.25M	Aggregate shipping cost
Freight-to-Price Ratio	16.57%	Freight cost as a share of item revenue
High-Installment Payment Share	30.65%	Payments split across 7+ installments
Multi-Category Order Rate	1.0%	Orders spanning more than one category

The headline finding: this marketplace's growth is more fragile than its top-line revenue trend suggests. A small seller base carries a disproportionate share of GMV, freight economics break down on cross-state shipments, and the platform is capturing almost none of the cross-sell opportunity implied by customers who do shop across categories over their lifetime.

Objectives

This analysis was scoped around four questions that go beyond standard delivery-and-satisfaction reporting:

- Quantify how dependent the marketplace is on its highest-revenue sellers, and where that dependency is riskiest (thin-coverage categories, single-seller regions).
- Measure the real cost of cross-state logistics — not just delivery speed, but freight economics relative to item price.
- Assess how exposed the platform is to deferred revenue through high-installment payment plans.
- Test whether the near-total absence of multi-category orders (1.0%) is a genuine merchandising gap or simply how this marketplace behaves, by comparing single-order behavior to lifetime customer behavior.

Each objective maps to a SQL workstream, a Python statistical validation step, and a dedicated Power BI dashboard, so every conclusion in this report traces back to a reproducible query and a specific visual.

Dataset Description

The dataset is the public Olist Brazilian e-commerce dataset: nine relational tables covering customers, orders, order line items, payments, reviews, products, sellers, and geolocation. Full column-level definitions and data-quality notes are documented separately in Data_Directory.md; the schema below shows how the tables connect.

Table	Rows	Grain
customers	99,441	One row per order-level customer record
orders	99,441	One row per order
order_items	112,650	One row per line item within an order
order_payments	103,886	One row per payment record (orders can have multiple)
order_reviews	99,224	One row per review
location	738,332 (post-dedup)	One row per geolocation point per zip prefix
products	32,951	One row per product
sellers	3,095	One row per seller
product_category_name	71	Category name lookup / translation

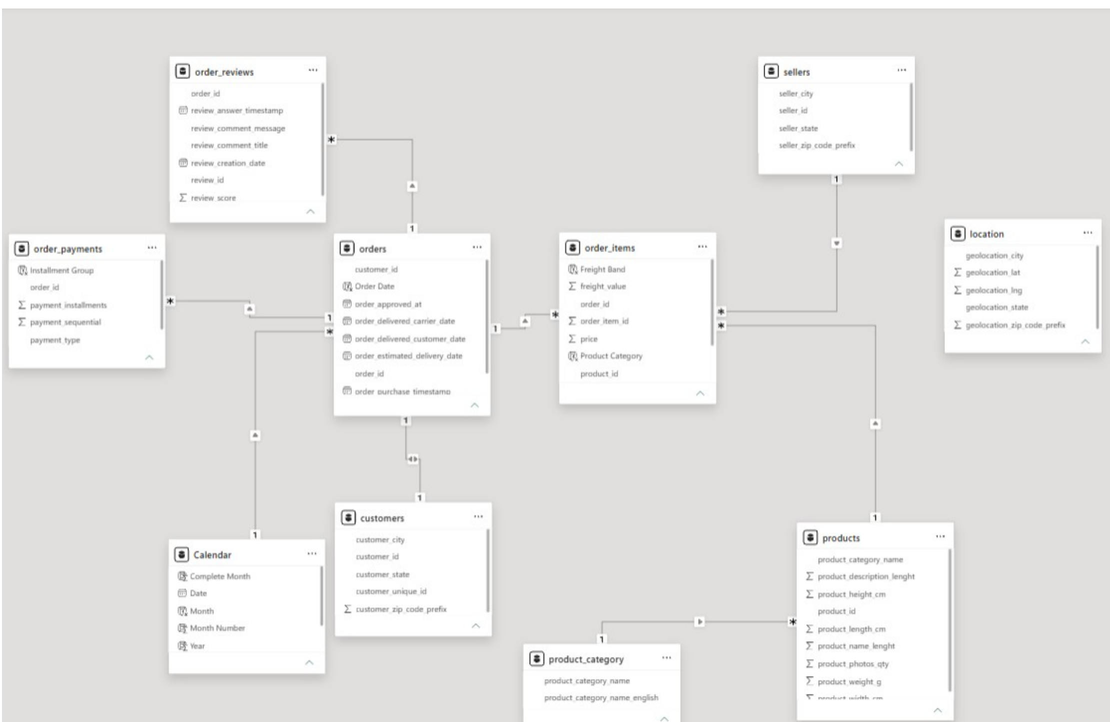


Figure 1. Relational data model — Power BI schema view

Data-quality steps applied before analysis: the location table's 261,831 full-row duplicates (26% of raw rows) were removed; date columns across orders and reviews were cast from text to datetime; and column names were trimmed of stray whitespace. Nulls in order_approved_at, order_delivered_carrier_date, and order_delivered_customer_date were left in place, since they reflect real order-lifecycle states (canceled, in-transit) rather than data-entry errors.

SQL Analysis

All source tables were loaded into SQL Server, and the analysis was built as twenty business questions in Business_Solution.sql, grouped into four workstreams that mirror the four dashboards below.

1. Seller Concentration & Dependency

- Ranked sellers by revenue and order volume to identify the platform's top earners.
- Used NTILE(10) to split sellers into revenue deciles and measure how much of total revenue each decile controls.
- Flagged product categories with the fewest distinct sellers — the categories most exposed to single-seller dependency.

2. Freight & Logistics Risk

- Compared average freight cost for same-state versus cross-state shipments.
- Calculated the share of order items where freight cost exceeds 50% of item price.
- Ranked sellers and state pairs by freight-to-price ratio to locate the least efficient shipping lanes.

3. Payment & Installment Risk

- Profiled order volume and payment value by installment count.
- Grouped installments into bands (1–3, 4–6, 7–12, 13+) to quantify how much payment value sits in long installment plans.
- Cross-referenced installment usage against product category to see where deferred payment is most common.

4. Cross-Sell & Basket Composition

- Counted distinct categories per order versus distinct categories per customer's full purchase history.
- Identified customers who buy across 3+ categories over their lifetime but rarely combine categories in a single order — the clearest evidence of an untapped cross-sell opportunity.
- Built a category-pair co-occurrence query to surface which category combinations already show the strongest cross-sell signal.

Python Exploratory Data Analysis

Before statistical testing, each table was profiled for shape, column types, nulls, duplicates, and key cardinality (Exploratory_Data_Analysis.ipynb). Key findings that shaped the rest of the analysis:

- customer_id has 99,441 unique values but customer_unique_id has only 96,096 — confirming customer_id is order-scoped and customer_unique_id is the correct key for any lifetime-behavior analysis (used directly in the cross-sell workstream).
- The location table carried 261,831 duplicate rows (26% of the raw file) — removed prior to any distance or zip-level aggregation.
- Date fields across orders and reviews were stored as text and required explicit datetime casting before any time-series or delivery-window calculation.
- No duplicate rows were found in customers, orders, order_items, order_payments, order_reviews, products, or sellers — the duplication issue was isolated to the geolocation table.

Cleaned tables were loaded into SQL Server as the single source of truth for both the dashboard layer and the statistical analysis notebook, so every number in this report and every dashboard KPI trace back to the same cleaned dataset.

Statistical Analysis

Descriptive Analysis

Business objective: establish a baseline for product prices, freight costs, payment values, installment usage, and reviews.

Metric	Mean	Median	Std Dev	Min	Max
Price	120.65	74.99	183.63	0.85	6,735.00
Freight Value	19.99	16.26	15.81	0.00	409.68
Payment Value	154.10	100.00	217.49	0.00	13,664.08
Installments	2.85	1.00	2.69	0	24
Review Score	4.09	5.00	1.35	1	5

The mean sits well above the median for price, freight, and payment value in every case — a small number of high-value transactions is pulling marketplace averages upward. This single pattern shows up again in every workstream below.

Distribution Analysis

Business objective: identify skewness in price, freight, and payment value to determine whether marketplace averages are being distorted by a small number of high-value transactions.

Histograms and boxplots for all three variables confirm strong right-skew: most transactions cluster at the low end of the range, while a long tail of high-value transactions stretches each distribution. Boxplots show a substantial number of points beyond the upper quartile whisker — these were not removed, since they represent legitimate high-value transactions rather than data-entry errors.

Outlier Analysis (IQR Method)

Business objective: identify unusually high prices, freight costs, and payment values that could represent financial or logistics risk.

Variable	Upper IQR Limit	Outlier Count	Share of Rows
Price	277.40	8,427	7.5%
Freight Value	33.25	12,134	10.8%
Payment Value	344.41	7,981	7.7%

The highest observed values — a $6,735 product, a $409.68 freight charge, and a $13,664.08 payment — all check out as coherent, legitimate transactions rather than entry errors, so they were retained rather than capped or removed.

Correlation Analysis

Business objective: measure the strength of relationships between price, freight cost, payment value, and installment usage.

- Price vs. Freight Value: r = 0.414 — a moderate positive relationship. Higher-priced items tend to cost more to ship, but price alone doesn't fully explain freight variation.

- Payment Value vs. Installment Count: r = 0.331 — a weak-to-moderate positive relationship. Higher-value payments lean toward more installments, but other factors are clearly at play too.

Hypothesis Testing

Business objective: confirm statistically — not just visually — whether cross-state shipments genuinely cost more to ship than same-state shipments, since this claim underpins the freight-risk recommendation later in this report.

Welch's two-sample t-test was run on freight_value, comparing same-state orders (n = 40,756, mean = $13.46) against cross-state orders (n = 71,894, mean = $23.69):

Test	t-statistic	p-value	Conclusion
Cross-state vs. same-state freight cost	130.72	< 0.001	Reject H_0 — difference is statistically significant

With a p-value far below 0.05, the ~76% higher average freight cost on cross-state orders is not sampling noise — it's a structural feature of how this marketplace ships goods, which directly supports Recommendation 2 below.

Confidence Interval Analysis

Business objective: estimate a reliable range for true average freight cost and payment value, rather than relying on a single point estimate.

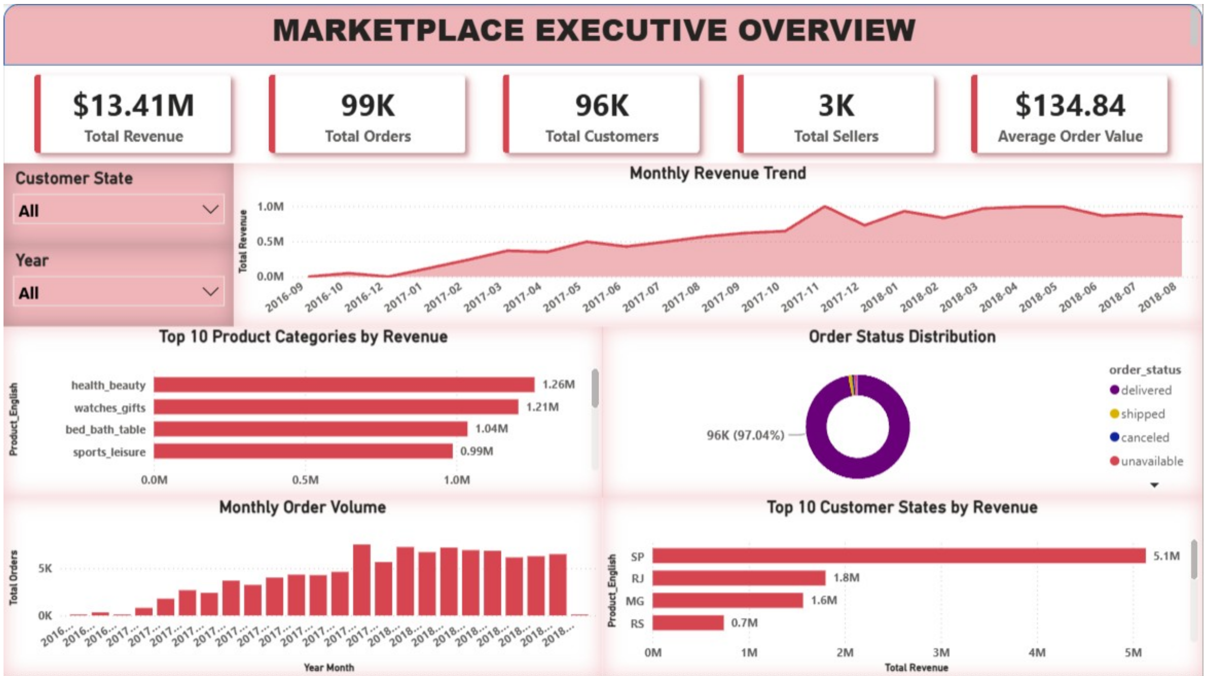
Metric	Sample Mean	95% CI Lower	95% CI Upper
Freight Value	$19.99	$19.90	$20.08
Payment Value	$154.10	$152.78	$155.42

Both intervals are narrow relative to their means — with close to 100,000 transactions behind each estimate, these averages are stable enough to use as ongoing baselines when evaluating future freight-policy or pricing changes.

Dashboard Summaries

Five interactive Power BI dashboards were built on top of the cleaned dataset, each corresponding to one analysis workstream. All dashboards share Year, State, and Category slicers for drill-down.

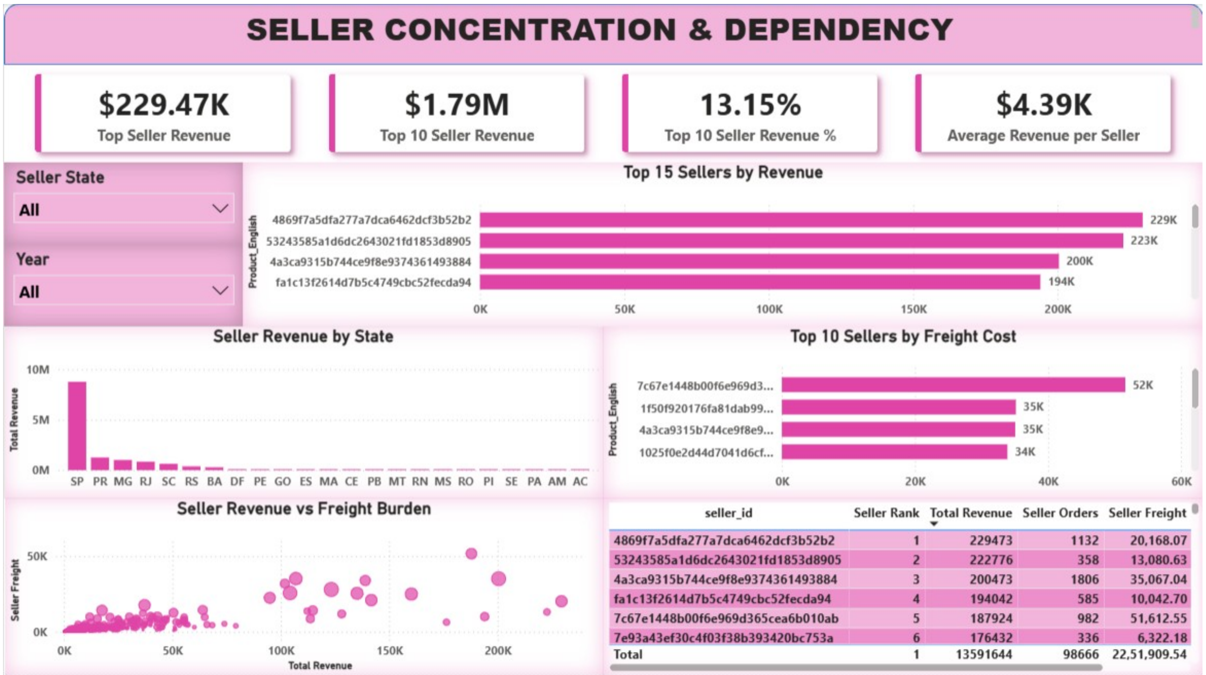
Marketplace Executive Overview



Marketplace Executive Overview — Power BI dashboard

- $13.41M in total revenue across 99K orders from 96K customers and 3K sellers, at a $134.84 average order value.
- 97.04% of orders reach delivered status — fulfillment reliability is not the marketplace's weak point.
- Monthly revenue climbed steadily from late 2016 through 2017 and plateaued through 2018 rather than continuing to compound — the growth story flattens right where the deeper risks in this report begin to show up.
- SP (São Paulo) dominates customer revenue by a wide margin over RJ, MG, and RS — the business is geographically concentrated on both the buy and sell side.

Seller Concentration & Dependency



seller_id	Seller Rank	Total Revenue	Seller Orders	Seller Freight
4869f7a5dfa277a7dca6462dcf3b52b2	1	229473	1132	20,168.07
53243585a1d6dc2643021fd1853d8905	2	222776	358	13,080.63
4a3ca9315b744ce9f8e9374361493884	3	200473	1806	35,067.04
fa1c13f2614d7b5c4749cbc52fecda94	4	194042	585	10,042.70
7c67e1448b00f6e969d365cea6b010ab	5	187924	982	51,612.55
7e93a43ef30c4f03f38b393420bc753a	6	176432	336	6,322.18
Total	1	13591644	98666	22,51,909.54

Seller Concentration & Dependency — Power BI dashboard

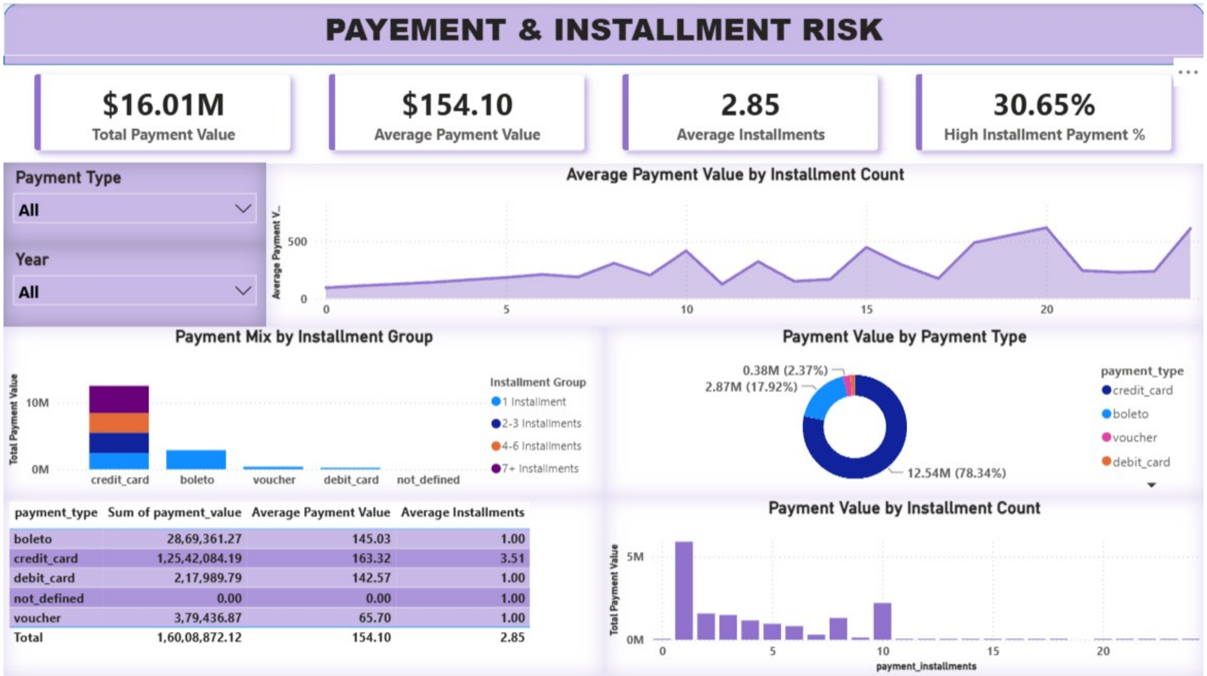
- The top 10 sellers alone generate $1.79M — 13.15% of total marketplace revenue — while the top decile of sellers (310 sellers) controls 67.5% of revenue.
- Average revenue per seller is $4.39K, dwarfed by the top seller's $229.47K — a small number of sellers are carrying the platform.
- Seller revenue is almost entirely sourced from SP-based sellers, compounding the geographic concentration seen on the Executive Overview.
- The revenue-vs-freight-burden scatter shows the highest-revenue sellers also carry the highest absolute freight cost — losing any one of them would hit both GMV and logistics capacity at once.

Freight & Logistics Risk

Freight & Logistics Risk — Power BI dashboard

- $2.25M in total freight cost, averaging $19.99 per item, with freight running at 16.57% of item price in aggregate.
- Cross-state shipments average roughly $23.69 in freight versus $13.46 for same-state shipments — confirmed statistically significant in the Hypothesis Testing section above.
- 10.62% of order items are classified as high-freight, and freight cost climbs in a tight linear band with item price, with a cluster of high-value outliers pulling the ratio up further.
- SP-based sellers account for the overwhelming majority of total freight cost ($279K of the visible top-3 total) — unsurprising given seller concentration, but it means freight risk and seller risk are the same risk, not two separate ones.

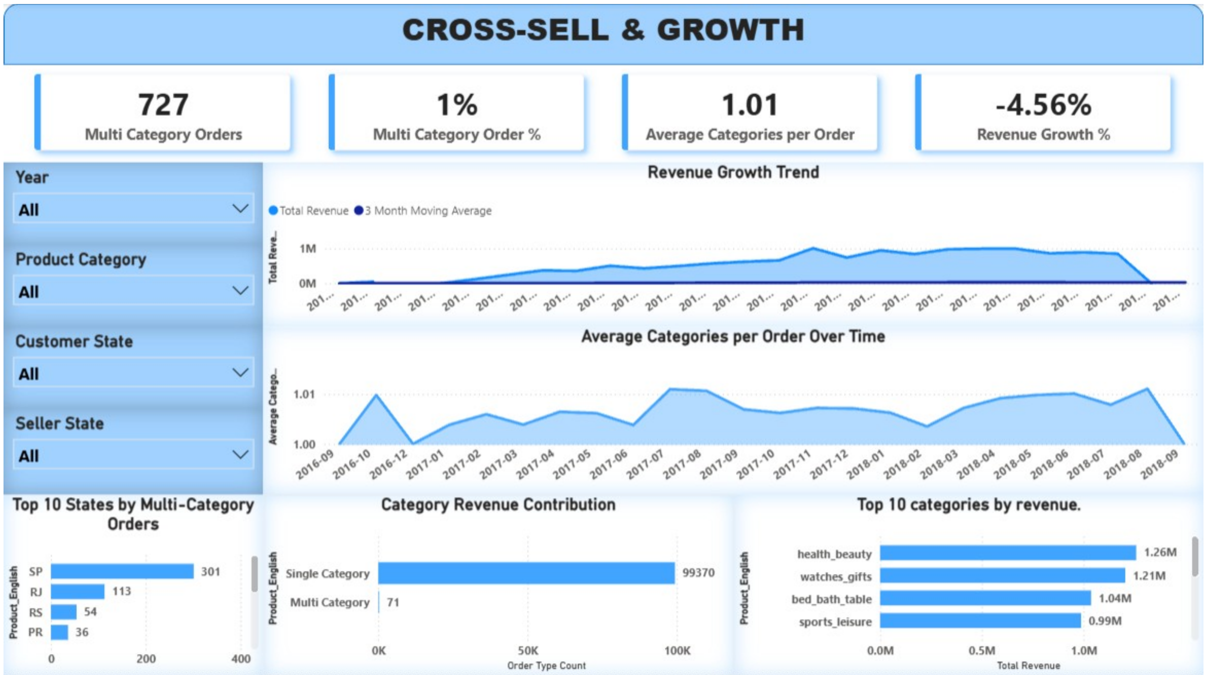
Payment & Installment Risk



Payment & Installment Risk — Power BI dashboard

- $16.01M in total payment value at an average of $154.10 per payment and 2.85 installments on average.
- 30.65% of payments use a high number of installments — nearly a third of transaction value is exposed to deferred, multi-period payment plans.
- Credit card is the dominant payment method at 78.34% of payment value ($12.54M); boleto (Brazil's cash-voucher payment method) accounts for a further 17.92%.
- Average payment value rises with installment count — customers making larger purchases are the ones most likely to spread payment out, concentrating deferred-revenue risk in the marketplace's highest-value transactions.

Cross-Sell & Growth



Cross-Sell & Growth — Power BI dashboard

- Only 727 orders (1.0%) span more than one product category, against 99,370 single-category orders — average categories per order sits at just 1.01.
- Revenue growth is currently −4.56% — the plateau visible on the Executive Overview has turned into a slight contraction in the most recent period.
- SP, RJ, RS, and PR lead in multi-category order count — the same states that dominate revenue also show what little cross-category behavior exists, rather than cross-sell emerging in new geographies.
- health_beauty, watches_gifts, bed_bath_table, and sports_leisure are the top revenue categories — useful starting points for testing bundling or recommendation strategies given how little natural cross-sell is happening today.

Key Business Insights

- Revenue is structurally concentrated, not just "top-heavy." 67.5% of revenue sitting with 10% of sellers, combined with the fact that the highest-revenue sellers also carry the highest freight burden, means seller attrition risk and logistics risk are entangled — losing a top seller doesn't just cut revenue, it destabilizes shipping economics too.
- Cross-state freight cost is a confirmed, statistically significant problem, not a perception. The ~76% freight premium on cross-state orders ($p < 0.001$) is large enough to be a genuine pricing or fulfillment-strategy lever, not noise to be smoothed over in reporting.
- Nearly a third of payment value carries deferred-revenue risk. A 30.65% high-installment share means marketplace cash flow is meaningfully out of sync with recognized revenue — worth tracking as its own risk metric, not just a payment-method footnote.
- Cross-sell is close to nonexistent at the order level (1.0%) despite categories like health_beauty and watches_gifts leading revenue independently — the SQL workstream on lifetime category behavior is the

next step to determine whether this is an unmet merchandising opportunity or simply normal single-purpose shopping behavior on this platform.

- Revenue growth has gone from a steady 2017 climb to a –4.56% contraction — this is the most urgent finding in the report, because it means the seller-concentration and freight-inefficiency risks above are no longer theoretical; they're already showing up in the top line.

Business Recommendations

1. Build a seller-diversification program for thin-coverage categories

Use the seller-count-by-category query from the SQL workstream to identify categories with the fewest active sellers, and prioritize seller recruitment there first — that's where a single seller's exit would do the most damage.

2. Introduce distance-aware freight pricing

Given the confirmed, statistically significant cross-state freight premium, evaluate a freight pricing or subsidy model that reflects actual seller-to-customer distance rather than a flat structure, and consider incentivizing sellers to stock in multiple regional hubs to cut cross-state shipment volume.

3. Monitor installment exposure as a cash-flow risk metric

Add the high-installment-share KPI (currently 30.65%) to regular financial reporting, not just the payments dashboard, so deferred-revenue exposure is visible to finance and not only to whoever is looking at the Power BI dashboard that week.

4. Pilot targeted bundling using the lifetime cross-sell signal

Rather than assuming the 1.0% multi-category order rate is the ceiling, use the customers-who-buy-3+-categories-but-never-combine-them query to build a targeted test: recommend a second category to those specific customers at checkout and measure whether basket size moves.

5. Treat the recent revenue contraction as the top-priority follow-up

The –4.56% growth figure deserves its own root-cause analysis — cut it by seller cohort, state, and category to determine whether it's concentrated among the top-decile sellers (which would validate the concentration-risk thesis directly) or spread evenly across the marketplace.

Future Improvements

- Extend the freight analysis with the geolocation table's lat/long coordinates to calculate actual seller-to-customer distance, rather than using same-state/cross-state as a proxy — the location table is currently underused relative to its size (738K rows post-dedup).
- Build a proper time-series forecast (e.g. SARIMA or Prophet) to replace the 3-month moving-average baseline used in this analysis, particularly given the recent revenue contraction.
- Layer review-text sentiment analysis on top of review_score to see whether seller-concentration or freight-cost patterns show up in what customers actually write, not just the numeric score.
- Formalize the cross-sell hypothesis test with a proper affinity/association-rule analysis (e.g. Apriori) on the category co-occurrence query, rather than relying on descriptive counts alone.
- Automate the seller-concentration and freight-risk KPIs as a scheduled refresh with alerting thresholds, so a shift like the current revenue contraction is flagged before it shows up a quarter later in an executive review.

Conclusion

This marketplace is not struggling with the problem most public analyses of this dataset focus on — delivery reliability is strong at 97.04%. The real risk sits one level up the stack: a small group of sellers carries most of the revenue and most of the freight burden simultaneously, cross-state shipping is measurably and significantly more expensive, nearly a third of payment value is deferred through installments, and the platform is capturing almost none of its apparent cross-sell potential. None of these are fulfillment problems — they're commercial structure problems, and the recent shift from steady growth to a –4.56% contraction suggests they've already started to bite. The recommendations in this report are sequenced to address the most entangled risk first (seller concentration and freight, which move together) before tackling the more exploratory opportunity (cross-sell), on the view that stabilizing the base is the precondition for growth, not a parallel track to it.